WORKED SOLUTION

STEP II 1993, Question 11

Mechanics • Fluid dynamics

STEP II 1993, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

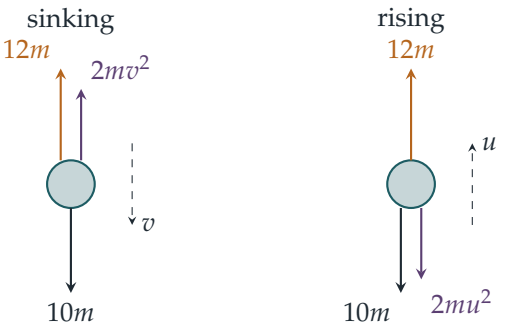
step.maths.org/questions/93-s2-q11

The key idea

Once the body is in the liquid, the buoyancy beats gravity, so there is a constant net upward force as well as the v^2 resistance. The resistance always opposes the motion, so its direction flips when the body turns round, and the sinking and rising phases need separate equations of motion. Each one separates neatly. Use $\frac{\mathrm{d}v}{\mathrm{d}t}$ when we want times and $v\frac{\mathrm{d}v}{\mathrm{d}x}$ when we want distances.

Solution

We model the body as a particle. In the liquid its weight is $10m$ N downwards and the buoyancy is $\frac{6}{5} \times 10m = 12m$ N upwards, so together they give a net force of $2m$ N *upwards*, whichever way the body is moving. The resistance $2mv^2$ N acts against the velocity.



The sinking phase

Take downwards as positive, with x the depth below the surface and v the downward speed. Newton's second law gives

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = 10m - 12m - 2mv^2 \qquad \Longrightarrow \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = -2(1+v^2).$$

Suppose the body enters with speed $V > 0$. Separating the variables and integrating from entry ($t = 0$, $v = V$) to the moment it stops sinking ($t = T_1$, $v = 0$):

$$\int_V^0 \frac{\mathrm{d}v}{1+v^2} = -2\int_0^{T_1} \mathrm{d}t \qquad \Longrightarrow \qquad T_1 = \tfrac{1}{2}\arctan V.$$

Since $\arctan V < \frac{\pi}{2}$ for every finite V, we get $T_1 < \frac{1}{4}\pi$: the body stops sinking less than $\frac{1}{4}\pi$ seconds after entering, however fast it was dropped in. (Also $\frac{\mathrm{d}v}{\mathrm{d}t} \le -2$ throughout, so it certainly does stop.)

Depth reached when $V = 1$

Now $T_1 = \frac{1}{2}\arctan 1 = \frac{\pi}{8}$. For the depth, write the acceleration as $v\,\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = -2(1+v^2) \qquad \Longrightarrow \qquad \int_1^0 \frac{v\,\mathrm{d}v}{1+v^2} = -2\int_0^D \mathrm{d}x,$$

so $-\frac{1}{2}\ln 2 = -2D$, and the greatest depth is

$$D = \tfrac{1}{4}\ln 2 \text{ metres},$$

as required.

The rising phase

At depth D the body is momentarily at rest, but the net upward force of $2m$ N is still there, so it starts to rise. Now the resistance acts downwards. Take upwards as positive, with y the height risen from the lowest point and u the upward speed:

$$m\frac{\mathrm{d}u}{\mathrm{d}t} = 12m - 10m - 2mu^2 \qquad \Longrightarrow \qquad \frac{\mathrm{d}u}{\mathrm{d}t} = 2(1-u^2).$$

Ask yourself Why can't we keep the sinking equation and just let v go negative?

The term $-2mv^2$ always points upwards, whatever the sign of v. Once the body is rising, the resistance must point downwards, so the equation changes. (A single equation would need $-2mv|v|$.) Notice too that $u < 1$ throughout the ascent: u starts at 0 and $\frac{\mathrm{d}u}{\mathrm{d}t} > 0$ only while $u < 1$, so u can only approach the terminal speed 1 ms^{-1}. This keeps $1 - u^2$ positive below.

Speed at the surface. Using $u\,\frac{\mathrm{d}u}{\mathrm{d}y} = 2(1-u^2)$ and integrating from $y = 0$, $u = 0$ to $y = D$, $u = U$:

$$\int_0^U \frac{u\,\mathrm{d}u}{1-u^2} = 2\int_0^D \mathrm{d}y \qquad \Longrightarrow \qquad -\tfrac{1}{2}\ln(1-U^2) = 2D = \tfrac{1}{2}\ln 2.$$

So $1 - U^2 = \frac{1}{2}$, and the body returns to the surface with speed $U = \dfrac{1}{\sqrt{2}}\ \mathrm{ms}^{-1}$.

Time to rise. Using $\frac{\mathrm{d}u}{\mathrm{d}t} = 2(1-u^2)$ with partial fractions $\dfrac{1}{1-u^2} = \dfrac{1}{2}\left(\dfrac{1}{1+u} + \dfrac{1}{1-u}\right)$:

$$T_2 = \frac{1}{2}\int_0^U \frac{\mathrm{d}u}{1-u^2} = \frac{1}{4}\left[\ln\frac{1+u}{1-u}\right]_0^U = \frac{1}{4}\ln\frac{1+\frac{1}{\sqrt{2}}}{1-\frac{1}{\sqrt{2}}} = \frac{1}{4}\ln\frac{\sqrt{2}+1}{\sqrt{2}-1},$$

multiplying top and bottom by $\sqrt{2}$ in the last step.

Total time. The body reaches the surface again at

$$T_1 + T_2 = \frac{\pi}{8} + \frac{1}{4}\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right) \text{ seconds}$$

after entering the pool, as required.

Checks

- Since $\dfrac{\sqrt{2}+1}{\sqrt{2}-1} = (\sqrt{2}+1)^2$, the rising time is $T_2 = \frac{1}{2}\ln(1+\sqrt{2}) \approx 0.4407\,\mathrm{s}$, and the total is $\approx 0.3927 + 0.4407 = 0.8334\,\mathrm{s}$.
- Stepping the motion numerically (Runge–Kutta, with the resistance written as $-2v|v|$) gives a stopping time $0.39270\,\mathrm{s}$, a depth $0.17329\,\mathrm{m} = \frac{1}{4}\ln 2$, a return speed $0.70711\ \mathrm{ms}^{-1}$ and a return time $0.83339\,\mathrm{s}$, all matching.
- The return speed is less than the entry speed, as it must be: the resistance does negative work on both legs.

Common slips

- Keeping the resistance upwards on the way back up, which would give a constant net upward force of $2m + 2mu^2$ and a much faster rise.
- Forgetting that the buoyancy outweighs gravity, so the net force is $-2m$ (downward positive), not $+2m$.
- Using $\frac{\mathrm{d}v}{\mathrm{d}t}$ for the depth or $v\frac{\mathrm{d}v}{\mathrm{d}x}$ for the times, and then getting stuck on the integral.

Beyond STEP

Quadratic drag makes the equation of motion nonlinear, yet it is still separable, which is why arctan appears going down and $\tanh^{-1}$ going up (indeed $T_2 = \frac{1}{2}\tanh^{-1} U$). The ascent approaches a terminal speed of $1\ \mathrm{ms}^{-1}$, where drag balances the net upward force. A first-year course treats such models systematically, including the phase-line picture of an equation like $\dot{u} = 2(1-u^2)$. Kleppner and Kolenkow, *An Introduction*

to Mechanics, covers velocity-dependent forces, and Boyce and DiPrima, *Elementary Differential Equations*, covers separable and autonomous equations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.